

Pharmaceutical Report

Group 22

Executive Summary

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The pharmaceutical data is a repeated cross sectional data frame showing how medicine was distributed in different regions over different years.

Note that just because medicine was dispensed does not mean that it was taken.

Additional district level variables to look into:

Income, age, employment, education, racial profile (Probably not), family dependencies.

Ethics, Privacy and Security

- Ethical considerations:

- Any comparison of groups being done in this project shouldn't cause have any implications about financial status between said groups. For example, Māori people take less medication than Pākehā because they can't afford them.
- Given the nature of the data collection, no causal conclusions can be drawn. The analysis will only be able to create commentary about the trends and relationships between the variables.

- Privacy considerations:

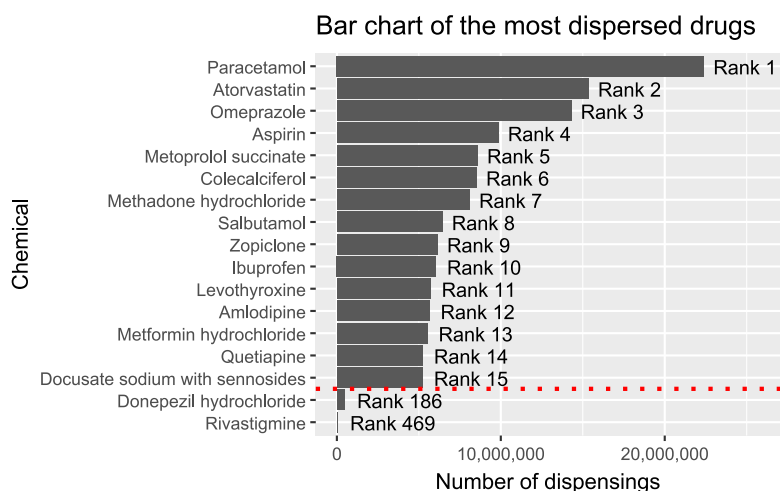
- We should not be removing the suppression layer from the count of medication distribution. While there is an oversight in how this data is provided that allows users to do this, and there

is code demonstrating that this is possible in this project, the values were randomly adjusted so that for the example instance doesn't have the true count revealed.

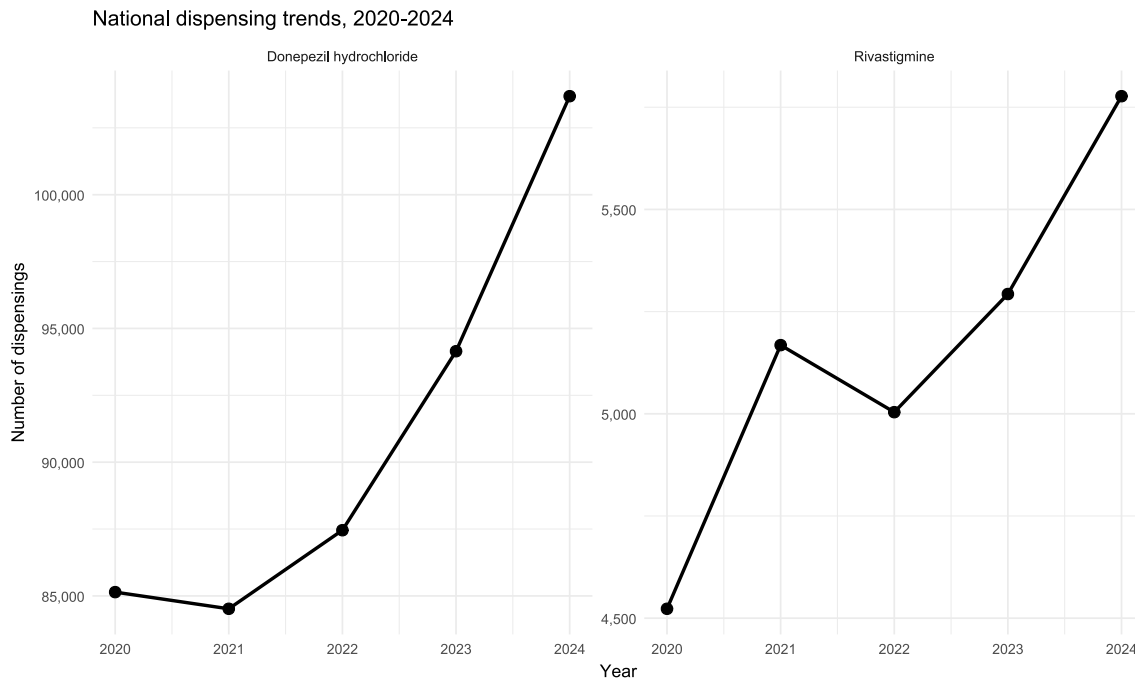
- For all analysis, the suppressed values will be imputed. Currently the method of replacing these with the median of the suppression range.

Exploratory Data Analysis

The graph below shows an unequal distribution of dispensing of medications of the 15 most widely distributed chemicals. Paracetamol is by far the most dispensed, with over double the amount of dispensings compared to the aspirin, the chemical in 4th place. The second and third most widely distributed medications are atorvastatin and omeprazole. Overall, the distribution has a long tail. Donepezil hydrochloride and Rivastigmine aren't highly distributed enough to make the top 15, with the former ranking 186th across the 5 years of data, and the latter ranking 469th over the same period.



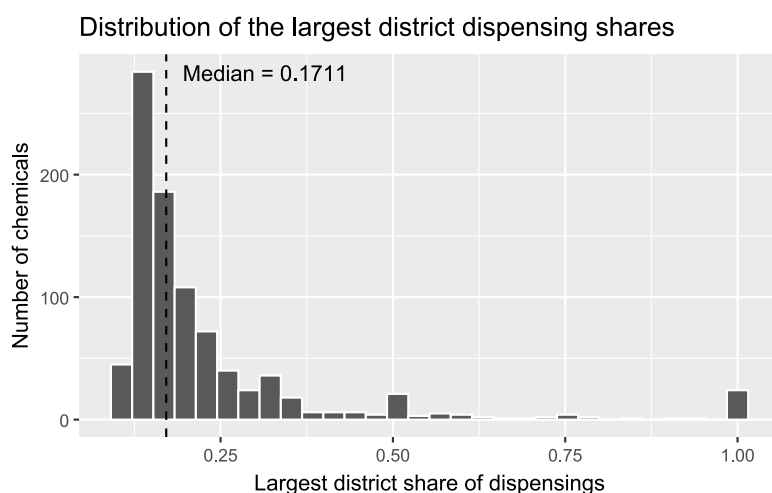
Looking at the line charts below shows the dispensing count for the two primary dementia medications annually from 2020 to 2024. Both show an upward trend of distributions over time.



The table below shows the same information in table format, broken down by count and ranks of the two dementia medications of interest. There has been an upwards trend in dispensing for both of these medications, while rivastigmine has had its rank reduced in recent years. This is because while the dispensing of this medication has increased in these years, not as much as other medication that had a lower dispensing count that rivastigmine in 2021.

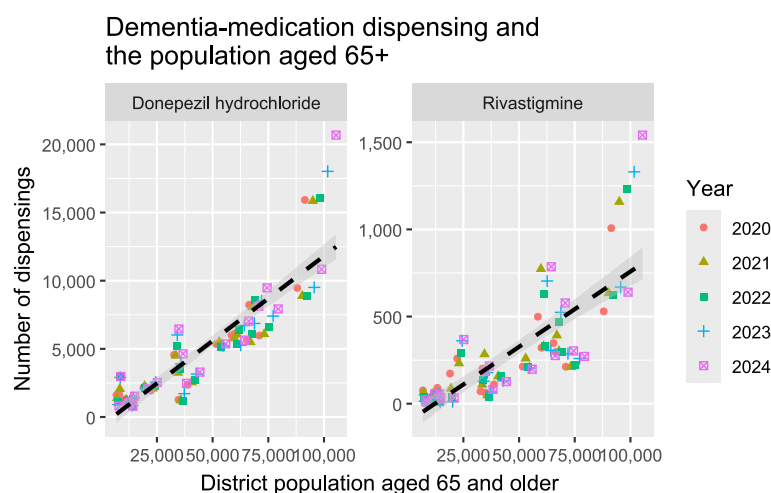
Year	DH count	R count	DH rank	R rank
2020	85142	4523	188	466
2021	84519	5168	189	454
2022	87456	5004	192	457
2023	94148	5293	183	467
2024	103683	5777	179	474

It is useful to know how frequently medications are distributed in a high concentration to a few health districts relative to an even spread across all health districts. One way to measure this is by using the proportion of dispensings going toward the health district where most of the dispensings occur. There are 19 health districts, and if the dispensings occurred equally across said districts, this metric would take the value of $1/19$. In the opposite extreme, if the medication was distributed in only 1 district, the metric would take the value of 1. Below is a histogram showing how each of the medications fall on this proportion scale.

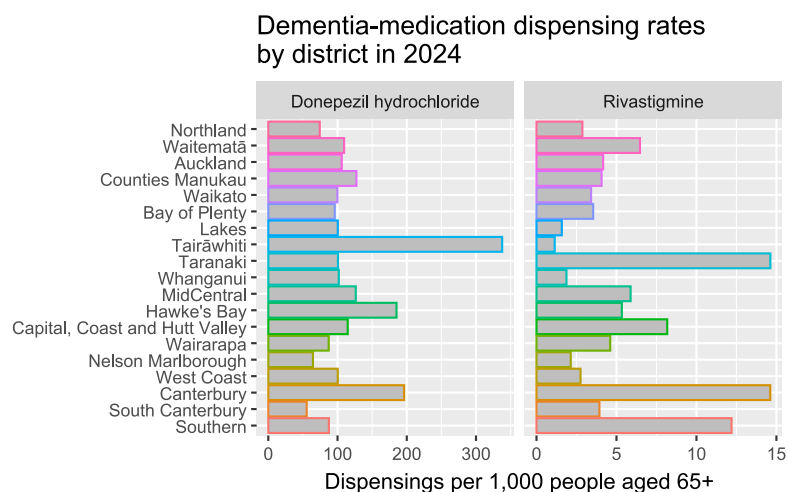


The median value is 0.1711, meaning that a typical medication will have roughly 17% of their medication go towards the district that leads in dispensing. The lowest observed proportion is 0.106283 meaning that no medication is equally distributed across all districts. There's a small group of medications that have a proportion value of 1, meaning that they are only distributed in a single district. Note that this metric doesn't speak to the number of dispensings for these medications, and are likely very niche with an estimated mean of 7.38 and max of 55. The medications with these lower dispensings may be very niche, available from retailers outside of pharmacies or not recorded in some districts, but this is speculative.

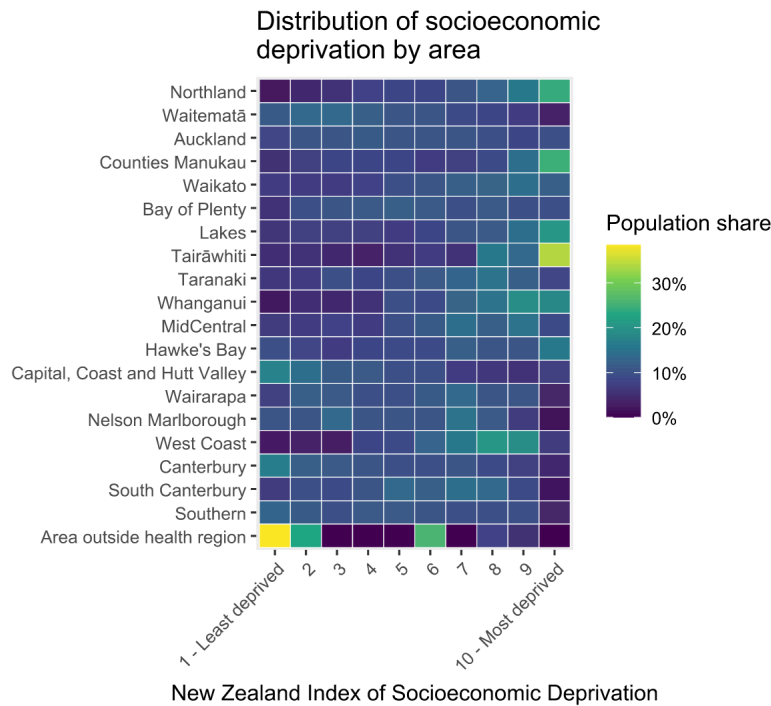
Dementia as a risk is strongly associated with old age. This makes the districts with a higher count of aged population more relevant to this analysis than total population size in the health district. The graphs below show a positive relationship between aged population count and dispensings of the two key dementia medications. This relationship is stronger for Donepezil Hydrochloride, but both show that there are some observations above the line of best fit on the higher end of the district aged population. These observations belong to the Canterbury district, and as the years increase it can be seen diverging further away from the line of best fit. Overall, this high correlation means that the older population size accounts for a large amount of the inter-district variation.



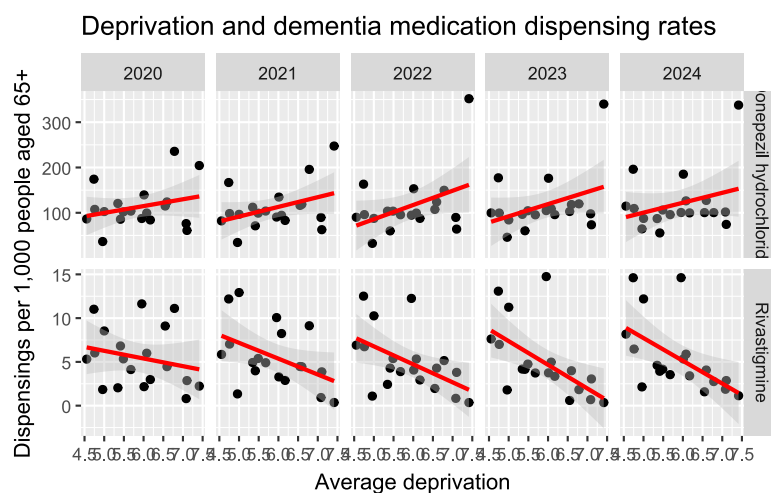
Crude rates were calculated as annual dispensings per 1,000 people aged 65 years and over for the 2024 year. The table below shows the variation in these dispensing rates across the different districts.



The first observation of note is that donepezil hydrochloride has a much higher rate than rivastigmine. Tairāwhiti has the highest overall rate when you take into account said larger rates for donepezil hydrochloride, as it is the district that has the highest rate for this medication. Canterbury and Hawke's Bay have the second and third highest rates respectively. Alternatively, Rivastigmine has the highest concentration of dispensing in Canterbury, Taranaki and Southern. Taranaki doesn't stand out for being high or low with donepezil hydrochloride, and Tairāwhiti has the lowest dispensing rate for rivastigmine. This could be related to the high proportion of the population of Tairāwhiti falling higher on the latest (2023) NZ deprivation scale as seen below.



The plots below show how the average dispensing rate per aged person changes for the two dementia medications over the deprivation index. For donepezil hydrochloride, the trend in the relationship between the dispensing rate and the deprivation index increases as time increases, peaking at 2022, followed by a decreasing trend in subsequent years. Alternatively, these trends have consistently steeper downward trends as the years increase. This is a strong argument to test interaction effects between year and deprivation index when predicting the number of medication dispensings.



These results remain exploratory. There are only about 19 districts per year, the same districts are observed repeatedly, and district deprivation changes little over the study period. Confidence intervals are wide, especially for rivastigmine, and the slopes may be sensitive to influential districts such as Tairāwhiti, Canterbury or Taranaki.